Seminar Announcement

Computer Engineering

Algorithmic Aspects of Throughput-Delay Performance for Fast data Collection in Sensor Networks

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10:00AM – 12:00PM
EEB 349

Abstract:
Convergecast, namely the many-to-one flow of data from a set of sources to a common sink over a tree-based routing topology, is a fundamental communication primitive in wireless sensor networks. For real-time, mission-critical, and high data-rate applications, it is often critical to maximize the aggregated data collection rate (throughput) at the sink node, as well as minimize the time (delay) required for packets to get there. In this thesis, we look into the algorithmic aspects of jointly optimizing both throughput and delay for aggregated data collection in sensor networks. Our contributions are in designing efficient algorithms with provably good, worst-case performance bounds for arbitrarily deployed networks. To the best of our knowledge, we are the first ones to address these two mutually conflicting performance objectives -- throughput and delay -- under the same optimization framework and develop techniques to meet the stringent requirements for fast data collection.

Our approach in addressing the throughput-delay performance trade-off comprises three techniques: (i) multi-channel scheduling, (ii) routing over optimal topologies, and (iii) transmission power control. We exploit the benefits of multiple frequency channels to design efficient TDMA scheduling algorithms, both under the graph-based and SINR-based interference models. In order to bound the maximum delay and further enhance the data collection rate, we study the degree-radius trade-off in spanning trees and propose bicriteria approximation algorithms to construct bounded-degree-minimum-radius spanning trees with constant factor guarantees on the maximum node degree as well as the tree radius. Lastly, to reduce interference and allow for more throughput, we design efficient, distributed power control schemes for sensor networks deployed in 3-D.

Defense Committee:

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